Practical Exercise: Google Big Query

Dataset: Retail Sales Dataset

```
## Q1: Filter all transactions that occurred in the year 2023
SELECT *
FROM `golden-bloom-493417-f3.RetailSales.SalesSales`
WHERE EXTRACT(YEAR FROM `Date`) = 2023;
```

Query completed

Query results

Chat | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	191	2023-10-18	CUST191	Male	64	Beauty	1	25	25
2	204	2023-09-28	CUST204	Male	39	Beauty	1	25	25
3	230	2023-04-23	CUST230	Male	54	Beauty	1	25	25
4	232	2023-02-06	CUST232	Female	43	Beauty	1	25	25
5	309	2023-12-23	CUST309	Female	26	Beauty	1	25	25
6	310	2023-10-12	CUST310	Female	28	Beauty	1	25	25
7	363	2023-06-03	CUST363	Male	64	Beauty	1	25	25
8	371	2023-02-21	CUST371	Female	20	Beauty	1	25	25
9	397	2023-03-10	CUST397	Female	30	Beauty	1	25	25
10	454	2023-02-22	CUST454	Female	46	Beauty	1	25	25
11	512	2023-11-07	CUST512	Female	57	Beauty	1	25	25
12	739	2023-11-29	CUST739	Male	36	Beauty	1	25	25

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```
## Q2: Transactions where Total Amount > average Total Amount

SELECT *
FROM `golden-bloom-493417-f3.RetailSales.SalesSales`
WHERE `Total Amount` >
    (SELECT AVG(`Total Amount`)
    FROM `golden-bloom-493417-f3.RetailSales.SalesSales`);
```

This query will process 72.69 KB when run.

Using on-demand processing quota

Query results

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Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	21	2023-01-14	CUST021	Female	50	Beauty	1	500	500
2	28	2023-04-23	CUST028	Female	43	Beauty	1	500	500
3	128	2023-07-05	CUST128	Male	25	Beauty	1	500	500
4	220	2023-03-03	CUST220	Male	64	Beauty	1	500	500
5	238	2023-01-17	CUST238	Female	39	Beauty	1	500	500
6	364	2023-08-23	CUST364	Female	19	Beauty	1	500	500
7	408	2023-04-15	CUST408	Female	64	Beauty	1	500	500
8	537	2023-06-03	CUST537	Female	21	Beauty	1	500	500
9	539	2023-06-08	CUST539	Male	25	Beauty	1	500	500

Results per page: 50 | 1 – 50 of 350

```
## Q3: Calculate total revenue

SELECT
  SUM(`Total Amount`) AS Total_Revenue
FROM `golden-bloom-493417-f3.RetailSales.SalesSales`;
```

Query completed

Using on-demand processing quota

Query results

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Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Total_Revenue
1	456000

Results per page: 50 | 1 – 1 of 1

```sql
## Q4: Display distinct product categories
SELECT DISTINCT
  `Product Category`
FROM `golden-bloom-493417-f3.RetailSales.SalesSales`;
```

Query completed

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Query results

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Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Product Category
1	Beauty
2	Clothing
3	Electronics

Results per page: 50 | 1 – 3 of 3

```sql
## Q5: Total quantity sold per product category
SELECT
  `Product Category`,
  SUM(`Quantity`) AS Total_Quantity
FROM `golden-bloom-493417-f3.RetailSales.SalesSales`
GROUP BY `Product Category`;
```

This script will process 254.03 KB when run.

Using on-demand processing quota

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Product Category	Total_Quantity
1	Beauty	771
2	Clothing	894
3	Electronics	849

```sql
## Q6: Create Age_Group
SELECT
  `Customer ID`,
  `Age`,
  CASE
    WHEN `Age` < 30 THEN 'Youth'
    WHEN `Age` BETWEEN 30 AND 59 THEN 'Adult'
    ELSE 'Senior'
  END AS Age_Group
FROM `golden-bloom-493417-f3.RetailSales.SalesSales`;
```

Query completed

Using on-demand processing quota

Query results

Chat | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Customer ID	Age	Age_Group
1	CUST191	64	Senior
2	CUST204	39	Adult
3	CUST230	54	Adult
4	CUST232	43	Adult
5	CUST309	26	Youth
6	CUST310	28	Youth
7	CUST363	64	Senior
8	CUST371	20	Youth
9	CUST397	30	Adult

Results per page: 50 | 1 – 50 of 1000

```sql
## Q7: High-value transactions (>500) per gender
SELECT
  `Gender`,
  COUNTIF(`Total Amount` > 500) AS High_Value_Transactions
FROM `golden-bloom-493417-f3.RetailSales.SalesSales`
GROUP BY `Gender`;
```

Query completed

Using on-demand processing quota

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Gender	High_Value_Tran...
1	Male	144
2	Female	155

```sql
## Q8: Product categories with revenue > 5000
SELECT
  `Product Category`,
  SUM(`Total Amount`) AS Total_Revenue
FROM `golden-bloom-493417-f3.RetailSales.SalesSales`
GROUP BY `Product Category`
HAVING SUM(`Total Amount`) > 5000;
```

Query completed

Using on-demand processing quota

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Product Category	Total_Revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

```sql
## Q9: Unit cost category
SELECT
  `Transaction ID`,
  `Price per Unit`,
  CASE
    WHEN `Price per Unit` < 50 THEN 'Cheap'
    WHEN `Price per Unit` BETWEEN 50 AND 200 THEN 'Moderate'
    ELSE 'Expensive'
  END AS Unit_Cost_Category
FROM `golden-bloom-493417-f3.RetailSales.SalesSales`;
```

This script will process 318.9 KB when run.

Using on-demand processing quota

Query results

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Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Transaction ID	Price per Unit	Unit_Cost_Category
1	191	25	Cheap
2	204	25	Cheap
3	230	25	Cheap
4	232	25	Cheap
5	309	25	Cheap
6	310	25	Cheap
7	363	25	Cheap
8	371	25	Cheap
9	397	25	Cheap

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```sql
## Q10: Customers aged 40+ with spending level
SELECT
  `Customer ID`,
  `Age`,
  `Total Amount`,
  CASE
    WHEN `Total Amount` > 1000 THEN 'High'
    ELSE 'Low'
  END AS Spending_Level
FROM `golden-bloom-493417-f3.RetailSales.SalesSales`
WHERE `Age` >= 40;
```

Query completed

Using on-demand processing quota

Query results

Chat | Save results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Customer ID	Age	Total Amount	Spending_Level
1	CUST191	64	25	Low
2	CUST230	54	25	Low
3	CUST232	43	25	Low
4	CUST363	64	25	Low
5	CUST454	46	25	Low
6	CUST512	57	25	Low
7	CUST791	51	25	Low
8	CUST825	46	25	Low
9	CUST855	54	25	Low

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